

ENGINEERING ECONOMICS

(Syllabus - TU)

Year - III

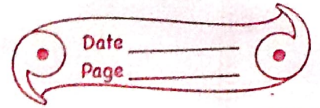
Part -

COURSE OBJECTIVE:-

After completing this course, students will be able to conduct simple economic studies. They will also be able to make evaluation of engineering projects & make decisions related to investment.

①. Introduction:-

- 1.1) Origin of Engineering Economy.
- 1.2) Principles of Engineering Economy.
- 1.3) Role of Engineers in Decision Making.
- 1.4) Cash Flow Diagram.

INTRODUCTION(1.1) ORIGIN OF ENGINEERING ECONOMY:-

→ Previously, in ^{the early} 19th Century, Engineers were mainly ~~considered~~ concerned with the design, construction & operation of machines, structures & processes.

→ Less attention was given to the use of resources (human & physical as well as capital & financial) that produced the final goods.

→ In the late 19th century, Engineers have been more interested in generating technological solutions, to make skillful financial analysis.

* → The study of Engineering Economy draws upon knowledge of engineering & economics to address the problems of allocating scarce resources.

→ Its definition suggests that the goal of engineers should transfer resources (natural) for the economic benefit of human race.

* → "The economic theory of location of Railways" written by A.M. Wellington in 1887 is known as pioneer ^{Author} engineering interest in economic evaluation.

↳ Wellington focussed that the capitalized cost method of analysis should be utilized in the selection of preferred lengths of rail lines.

* "Adam Smith" (father of economics) published "The wealth of nations".

* In 1920, C.L. Fish & O.B. Goldman looked at investment in engineering structures from the perspective of bond market & compound interest procedure for determining comparative economic values.

* In 1930, L. Grant published "Principles of Engineering Economy" & discussed about the importance of judgment factors, short-term investment, long-term investments in capital goods based on compound interest calculations. His many contribution resulted in the recognition that he is the father of engineering economy.
{Eugene L. Grant}

* ENGINEERING ECONOMICS:

→ Engineering economics is the application of economic techniques to the evaluation of design & engineering alternatives.

→ Engineering economics is concerned with the systematic evaluation of the costs & benefits of proposed technical & business projects.

→ The techniques & models of engineering economics assist people in making decisions.

→ The decisions are made by the engineers to choose one alternative over another & decisions often ~~reflect~~ are used to assess the economic merits of alternatives.

1.3 → Role of Engineers in Decision Making:-

→ The techniques & methodology of engineering economy can assist engineers in decision making process from the following grounds:-

- i) Understand the problem.
- ii) Define objective.
- iii) Collect relevant information/data.
- iv) Develop feasible alternatives.
- v) Identify the criteria for decision making.
- vi) Select the best alternative.
- vii) Evaluate each alternative.
- viii) Effective implementation & so on.

→ In decision making process, the role of engineers may also be required for different situations like:-

- i) Equipment & process selection.
- ii) Equipment replacement.
- iii) New product & production expansion.
- iv) Cost reduction.
- v) Capital budget allocation.
- vi) Service improvement.
- vii) Project evaluation etc.

Hence, the role of engineers is to assess the appropriateness of a given project, estimate its value & justify it from an engineering viewpoint.

1.2. PRINCIPLES OF ENGINEERING ECONOMY:-

→ The foundation for engineering economics is a **set of 7 principles** or fundamental concepts that provide a sound basis for development of engineering economics study.

→ These fundamental concepts are discussed in terms of following seven principles:-

①. Develop the alternatives:-

→ The feasible alternatives need to be identified & then defined for subsequent analysis.

→ A decision is made between atleast two or more alternatives.

→ Developing the alternatives is important because of the resulting impact on the quality of decision.

→ Remember, Quality derives from quantity.

②. Focus on difference:-

→ The expected differences in future outcomes among ~~time~~ the alternatives are relevant to their comparison & should be considered in the decision making.

→ If all the prospective outcomes of the feasible alternatives are exactly same, we would be indifferent & could make a decision using random selection.

→ Be very neutral & impartial between the alternatives.

③. Use a consistent viewpoint (economic viewpoint)

→ To determine prospective (future) outcomes of the feasible alternatives, the viewpoint should be consistently developed & defined.

→ Suppose all the alternatives have the same future cost to the public company. In this situation, the perspective of the employees & their satisfaction may be important decision criterion.

④. Use a common unit of measure:-

→ Common unit of measurement should be followed which will make easier the analysis & comparison.

→ For economic consequences, a monetary unit is a common measure.

→ You should translate other outcomes into the monetary units that will make subsequent analysis & comparison easier.

⑤. Consider all relevant criteria (social values & environmental aspect):-

→ Use of all relevant criteria is required for a decision making.

→ The primary criterion relates to provide maximum monetary return but other objectives like social, cultural, environmental justice etc should become the basis for additional criteria in the decision making process.

⑥. Make uncertainty explicit:- (stated clearly in detail)

→ Uncertainty should also be recognised in the analysis & comparison.

→ For example, change in competition, tax rates, prices of factor inputs (rent, wages, interest), political & natural disturbances etc.

→ Thus, dealing with uncertainty is an important aspect of engineering economic analysis.

(7) Revisit your decision:— (Self Evaluation):—

→ Improve decision results from an adaptive process.

→ For this, the projected outcomes of the selected alternative & actual results achieved should be subsequently compared.

→ Post evaluations of implemented decisions should be routinely done & the results used to future decision making.

→ Practice makes a man perfect.

* Engineering Economy Methodology & its Application:—

→ Methodology consists of the techniques & procedures from the technical content of the discipline.

→ Techniques incorporate:—

→ Seven principles (discussed above)
&

→ Tools used in engineering practice which are the topic for the remaining discussions.

→ The Applications ~~of~~ of engineering economy are diverse & found in most areas of an organization.

→ It is primarily concerned with the evaluation of alternatives uses of capital.

→ It is used to assist decision making in different situations.

→ The different situations where these methodologies are useful are as follows:—

(i) Equipment & process selection.

(ii) Equipment Replacement.

(iii) New product & production expansion.

(iv) Cost reduction.

(v) Capital Budget Allocation.

(vi) Service Improvement.

(vii) New project evaluation.

1.4 CASH FLOW & CASH FLOW DIAGRAMS:—

* Cash Flow:—

→ The analysis of economic events & transactions that affects on the cash position of a company is termed as Cash Flow.

→ It is a statement, showing the inflows & outflows of cash & cash equivalents during different time.

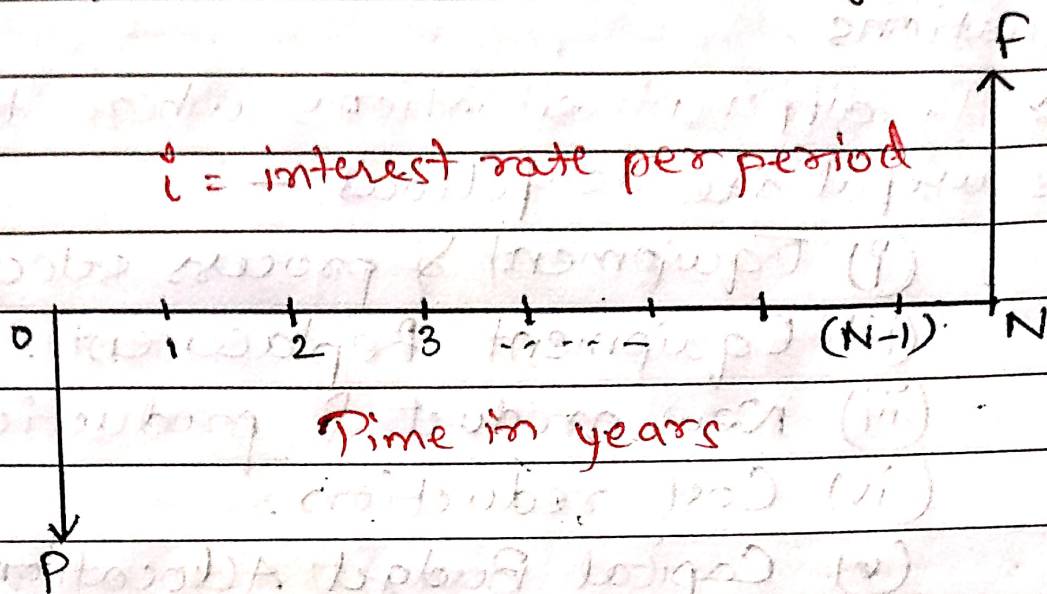
→ It is a statement of a company's ability to generate cash as well as their need of cash.

→ It is important because it is the basis for evaluating different alternatives.

→ Cash flow means actual rupees coming into or goes out during different time.

* Cash Flow Diagram: -

→ A typical cash flow for N years time scale:



In a cash flow diagram: -

(i) The horizontal line is a time scale with progression of time moving from left to right.

(ii) The period labels (on the horizontal line) are applied to the intervals of time.

i.e., the end of 1st period is coinciding with the beginning of 2nd period.

(iii) The Arrows signify cash flows: -

→ Downward arrow: - Cash Outflows.
(i.e., Expenses)

→ Upward arrow: - Cash Inflows
(i.e., Income & receipts)

Ex (1.1) An investment of Rs 10,000 can be made that will produce uniform revenue of Rs 5,310 for 5 years & then a positive salvage value of Rs 2,000 at the end of year 5. The annual expenses will be Rs 3,000 at the end of each year for operating & maintaining the project. Draw a cash flow diagram for the five year life of the project.

Soln:

Given,

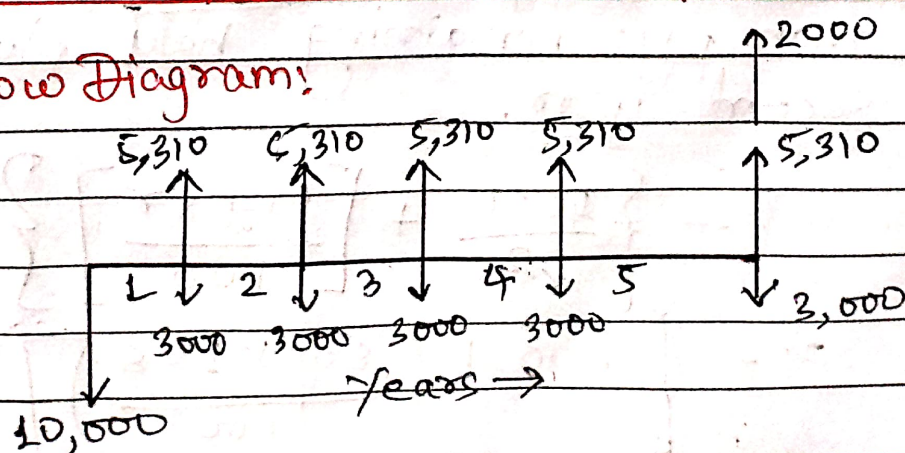
CASH
OUTFLOWS { Initial investment = Rs 10,000
Annual expenses = Rs 3,000 (at the end of each year)

CASH
INFLOWS { Revenue = Rs 5,310
Salvage value = Rs 2,000

Cash Flow Schedule: +

End of year	Cash Outflows (Rs)	Cash Inflows (Rs)
0	10,000	—
1	3,000	5,310
2	3,000	5,310
3	3,000	5,310
4	3,000	5,310
5	3,000	5,310 + 2,000

Cash Flow Diagram:



Ex. (1.2) A man borrowed Rs 1000 from a bank at 8% interest rate. Two end of year payments are done. At the end of first year, he will repay half of Rs 1000 principal plus the interest that is due. At the end of the second year, he will repay remaining half plus the interest for second year.

Draw CFD.

Soln:

Given:-

* At the end of 0th year / At the beginning of 1st year man borrowed money, so cash inflow of Rs 1000.

* At end of 1st year:-
man pays half of principle amount plus interest

$$= - \left\{ \frac{1000}{2} + \left[\frac{P \times t \times r}{100} \right] \right\} \quad \left[\begin{array}{l} \text{-ve sign for} \\ \text{cash outflow} \end{array} \right]$$

$$= - \left\{ 500 + \left[\frac{1000 \times 1 \times 8}{100} \right] \right\}$$

$$= - \{ 500 + 80 \}$$

$$= - \{ \text{Rs } 580 \} \quad (\text{Cash Outflow})$$

* At the end of 2nd year:-
man pays remaining half plus interest for second year.

$$= - \left\{ \frac{1000}{2} + \left[\frac{P \times t \times r}{100} \right] \right\}$$

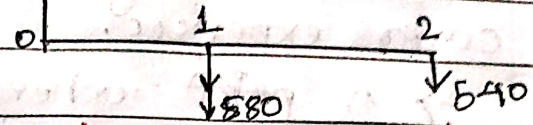
$$= - \left[500 + \left(\frac{500 \times 1 \times 8}{100} \right) \right]$$

$$= -540$$

$$= -\{125540\}$$

RS 1000

C.F.D.



* Essential Business Terminologies Used in Engineering Economic Analysis:-

(i) Annuity:-

→ A series of equal payments (amount of money) occurring at equal periods of time, like annually, monthly, weekly etc.

→ If paid at the end of each period → Annuity is known as ordinary or deferred annuity.

→ If paid at the beginning of each period → annuity is known as annuity due.

→ If an annuity goes on forever, then it is known as perpetuity.

(ii) Asset:-

→ Capital owned by a company.

→ An entity possessing market or exchange value & forming part of wealth.

* Real Asset:- land, building, plant, machinery, equipments etc.

* Financial Asset:- cash, shares, securities etc.

(iii) Break-evenpoint (BEP) :-

→ A point that indicates graphical representation of the relation between total revenue & total cost for that level of output (production) & sales at which revenue is just sufficient to cover all expenses.

→ The percentage of capacity operation of a manufacturing plant at which income will just cover expenses.

→ A point where the organization's financial status is neither profit nor loss.

(iv) Capital:-

→ A term describing wealth which may be utilized to economic advantage.

→ Man-made factors that contribute to the production of goods & services.

→ The non-human ingredients (factors) that may be utilized to economic advantage.

→ It includes cash, buildings, plants, equipment, raw materials, semi-finished & finished products, inventories etc.

→ Capital goods are used for production of other (final) goods.

Land

- Free gift of nature.
- Permanent factor of production.
- Fixed supply due to gift of nature.

Capital

- Manmade factor of production.
- Perishable factor of production.
- Flexible supply due to manmade.

Wealth

→ It may be idle property that may not generate income.

Capital

→ It is mobilized property which gives extra income like rent, output.

Money

→ It may be an un-invested stock amount that could not give any extra income.

Capital

→ It is an invested amount of money which creates extra income like interest or profit.

(v) Cash-Flow:-

→ The statement showing actual rupees passing into & out of the treasury of a financial venture.

Cash inflow: actual rupees coming into the firm.

Cash outflow: actual rupees going out of the firm.

(vi) Capital recovery (CR):-

→ Charging periodically to operation amounts that will ultimately equal to the amount of capital expenditure.

→ It is uniform annual cost of capital investment.

(vii) Compound Interest:-

→ That type of interest which is periodically added to the amount of principal investment (or loan) so that subsequent interest is based on the cumulative amount.

→ It includes interest on past interest.

(viii) Simple Interest:-

→ Interest is charged only on the basis of initial principal amount for all over the time period.

→ It is fixed % of the principal amount (loan) multiplied by the life of the loan.

(ix) Discount rate:-

→ The rate used to calculate the present value of future cash flows because of time preference. It is inverse of compounding.

(x) Depreciation:-

→ Declining value of capital assets due to wear & tear, passage of time, obsolescence etc.

→ Allocation of the cost of fixed asset over its economic useful life.

(xi) Free good:-

→ The goods obtained without any payment & human effort.

→ Free gift of nature.

(xii) Economic Good:-

→ A good that is scarce & having economic value.

→ Anything that is useful, transferable & not abundant.

(xiii) Economic Life (Useful Life):-

→ The length of a capital asset that will be economically useful.

→ The expected period that a property will be used in a trade or business to generate income.

(xiv) Economic efficiency:-

→ It is the ratio of the value of output obtained from an economic process to the value of input necessary to produce them.

(xv) Study Period:-

→ Expected length of time of any project to run & appraisal (assessment) of the result.

(xvi) Salvage Value:- [market/~~residual~~/residual] value:-

→ The value of capital asset at the end of economic life has to be taken into account.

→ The estimated value of a property at the end of its useful life.

(xvii) Decision making:-

→ A program of action undertaken as a result of established policy influencing the final outcome.

* Decision under certainty:-

→ A simple ~~analysis~~ ^{decision} that assumes complete information & no uncertainty is connected with analysis.

* Decision under risk:-

→ A decision for which the analyst elects to consider several possible futures where the probabilities of which can be estimated.

* Decision under uncertainty:- (uncertainty):-

→ A decision for which the analyst elects to consider several possible future where the probabilities of which cannot be estimated.

(xviii) Inflation:-

→ The situation of rapid rise in general price level & fall in the value of money.

(xix) Deflation:-

→ The situation of falling market price or rising value of money.

(xx) Utility:-

→ The power of satisfaction to a consumer by consuming goods or services.

(xxi) Opportunity Cost:-

→ It is the cost of the rejected opportunity to earn return from the use of scarce resources.
→ The value of second best opportunity foregone by deciding to do one thing rather than the other.

(xxii) Payback Period:-

→ Required time period (years/months) to recover the initial investment of the project.

(xxiii) Labor:-

→ The capacity of human effort (both mental & muscle) ~~to recover~~ available.

(xxiv) Demand:-

→ The various quantities of an item that a buyer is willing to buy at alternative prices, other things being equal.

(xxv) Supply:-

→ The various quantities of an item that a seller is willing to sell at an alternative prices, other things being equal.